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High-Level Design Document

Outbound Voice AI Collections Agent — “Maya” (Kapture Finance)

Version: 1.0 **Owner:** Voice AI Engineering **Scope:** Automated outbound collections call, identity-gated debt disclosure, PTP negotiation, disposition logging.

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1. Pipeline & Latency Budget

1.1 Architecture Flow

```
Telephony (SIP/PSTN)
     │
     ▼
Deepgram Nova-2 (STT)
     │
     ▼
Orchestrator / GPT-4o (LLM, temp 0.1)
     │
     ├──► Mock Webhook API (tool calls)
     │
     ▼
ElevenLabs / Cartesia (TTS)
     │
     ▼
Telephony Output
```

1.2 Latency Budget per Hop

Hop	Component	Target Latency	Notes
1	Network ingress (SIP → Vapi)	~50 ms	Carrier/PSTN dependent
2	STT (Deepgram Nova-2)	~200 ms	Streaming, partial transcripts used for barge-in
3	LLM first byte (GPT-4o)	~400 ms	Low temp (0.1) reduces retries/re-generation
4	Tool call round-trip (webhook)	~150–300 ms	Only on turns that trigger a function call
5	TTS synthesis first audio chunk	~300 ms	Streamed, not waiting for full

Hop	Component	Target Latency	Notes
			sentence
6	Network egress (Vapi → Telephony)	~150 ms	
Total (no tool call)		**< 1.2 s**	End-to-end target
Total (with tool call)		**< 1.5 s**	Acceptable bump on verification/PTP turns

1.3 Mitigations for Latency Overruns

- Stream TTS audio as soon as the first sentence is generated rather than waiting for the full LLM response.
- Keep the mock/production webhook synchronous and sub-300ms — no blocking I/O, no cold starts (keep server warm).
- Use `gpt-4o-mini` if `gpt-4o` first-byte latency exceeds budget under load.

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2. State Machine

2.1 States

State	Description
`INIT`	Call connects, greeting delivered
`AUTH_PENDING`	Awaiting identity verification input
`AUTHENTICATED`	`verify_customer` returned `verified: true`
`NEGOTIATION`	Debt disclosed, intent being determined (PTP / hardship / dispute / already paid)
`PTP_COLLECTED`	Promise-to-Pay captured and logged
`ESCALATED`	Routed to human agent (hardship or dispute)
`CALL_ENDED`	Disposition logged, call terminated

2.2 Transition Table

From	Event	To	Guard
`INIT`	Customer confirms identity ("yes, this is Rahul")	`AUTH_PENDING`	—
`INIT`	Customer denies / unavailable	`CALL_ENDED`	`mark_disposition (WRONG_PERSON)` fires first
`AUTH_PENDING`	`verify_customer → verified: true`	`AUTHENTICATED`	**Hard lock** — no debt term may be spoken before this

From	Event	To	Guard
			transition
AUTH_PENDING	verify_customer → verified: false	AUTH_PENDING (retry, max 2) then CALL_ENDED	After 2 failures, disposition NO_RESPONSE/escalation
AUTHENTICATED	Debt disclosed	NEGOTIATION	—
NEGOTIATION	Customer agrees to pay	PTP_COLLECTED	log_promise_to_pay + send_payment_link fire
NEGOTIATION	Customer claims already paid	CALL_ENDED	mark_disposition(ALREADY_PAID)
NEGOTIATION	Hardship claim	ESCALATED	escalate_to_agent(HARDSHIP_REQUEST)
NEGOTIATION	Dispute	ESCALATED	escalate_to_agent(DISPUTE)
NEGOTIATION / any	DNC request	CALL_ENDED	mark_disposition(DO_NOT_CALL) — immediate, no further negotiation
PTP_COLLECTED / ESCALATED	Wrap-up complete	CALL_ENDED	mark_disposition fires with final status

2.3 Explicit Compliance Rule

> Transitions out of AUTH_PENDING into AUTHENTICATED are strictly locked behind a successful verify_customer(status: success) tool response. The LLM must not use the words "overdue," "loan," "EMI," "amount," or "Kapture Finance debt" in any state prior to AUTHENTICATED.

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3. Intents & Entities Table

3.1 Intents

Intent	Triggers From State	Resulting Action
Confirm_Identity	INIT	→ AUTH_PENDING
Provide_Verification	AUTH_PENDING	Calls verify_customer
Promise_To_Pay	NEGOTIATION	Calls log_promise_to_pay, send_payment_link
Already_Paid	NEGOTIATION	Calls mark_disposition(ALRE

Intent	Triggers From State	Resulting Action
		ADY_PAID)
Hardship_Claim	NEGOTIATION	Calls escalate_to_agent(HARDSHIP_REQUEST)
Dispute_Debt	NEGOTIATION	Calls escalate_to_agent(DISPUTE)
Request_DNC	Any	Calls mark_disposition(DO_NOT_CALL), ends call
Wrong_Person	INIT	Calls mark_disposition(WRONG_PERSON), ends call

3.2 Entities

Entity	Type	Format	Example
PTP_Date	Date	ISO-8601	2026-08-14
PTP_Amount	Number	Decimal, INR	8499
Hardship_Reason	String	Free text	"job loss"
Verification_Code	String	4 digits or birth year	"1234" / "1995"

4. Tool / API Specifications

All tools return JSON matching Vapi's expected `{ results: [{ toolCallId, result }] }` shape (see `mock-server/server.js`). Full JSON Schemas are in `vapi/tool_definitions.json`.

Tool	Purpose	Key Inputs	Key Outputs
verify_customer	Authenticate caller before any disclosure	account_id, verification_code	verified (bool), message
log_promise_to_pay	Record PTP commitment	account_id, ptp_date, amount	success, ptp_id
send_payment_link	Dispatch payment link	account_id, channel (SMS/WhatsApp/BOTH)	success, message
escalate_to_agent	Route hardship/dispute to human	account_id, reason	success, queued
mark_disposition	Log final call	account_id,	success,

Tool	Purpose	Key Inputs	Key Outputs
	outcome	`status` (enum), `notes`	`disposition_logged`, `timestamp`

5. Auth & Data Safety Protocols

- **PII masking in logs:** Customer names are masked in server logs and analytics exports, e.g. `Rahul S****`. Full PAN/DOB values are never logged — only a boolean match result.
- **Zero-disclosure-before-auth:** The terms "Overdue," "Loan," "EMI," and "Kapture Finance debt" are blocked from the LLM's output vocabulary until `verify_customer` returns `verified: true`. This is enforced via the system prompt today; production should add a code-level output filter as a second layer of defense.
- **Verification code storage:** `verification_code` values are never persisted — they exist only in-memory for the duration of the tool call.
- **Transport security:** All webhook traffic runs over HTTPS (via ngrok in dev, a proper TLS cert in production).
- **Least privilege:** The mock server only exposes the 5 tool endpoints — no general data access.

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6. Compliance & Guardrails

- **RBI Fair Practices Code adherence:**
  - Calling window restricted to **08:00–19:00 local time** — enforced at the dialer/campaign-scheduling layer, not just the prompt.
  - No third-party debt disclosure — enforced by the `AUTH_PENDING` state lock.
  - Instant opt-out — any DNC/"stop calling" utterance is handled immediately, in any state, ahead of the normal negotiation flow.
- **Hallucination prevention:**
  - Temperature fixed at `0.1`.
  - The agent is explicitly instructed it **cannot offer unauthorized waivers greater than 10%** of the outstanding amount.
  - The agent cannot invent new dates, amounts, or account details — it only reflects values from `CUSTOMER & ACCOUNT CONTEXT` in the system prompt or from tool responses.
- **Tone constraint:** Calm, firm, supportive, respectful — no threats, no raised urgency language, no harassment patterns (aligns with fair collections norms).
- **Abusive-user handling:** One clear verbal warning, then a soft, polite hangup (see Edge Case Matrix below).

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7. Edge Cases Matrix

Edge Case	Detection	Handling
Abusive/hostile user	Profanity or threats detected in transcript	1 calm warning ("I understand this is frustrating, but I'll need to end the call if this continues") → soft hangup, `mark_disposition(status="NO_RESPONSE", notes="Abusive - terminated")`
Silent user / voicemail	No speech detected for N seconds	2 re-prompts ("Hello, are you still there?") → hangup with `mark_disposition(status="NO_RESPONSE")`
Mid-call language switch (English ↔ Hindi)	STT language confidence shift / code-switch detected	Prompt instructs agent to mirror the customer's language (Hindi/Hinglish) without losing state or previously captured entities
Wrong number	Customer confirms they are not Rahul Sharma and Rahul is unavailable	`mark_disposition(status="WRONG_PERSON")`, polite close, no debt terms ever used
Repeated failed verification	2 consecutive `verify_customer` failures	Do not disclose debt; offer to call back when the account holder is available; `mark_disposition(status="NO_RESPONSE", notes="Verification failed x2")`
Customer requests DNC mid-negotiation	"Stop calling me" / "put me on do not call"	Immediate `mark_disposition(status="DO_NOT_CALL")`, skip remaining negotiation, end call
Partial/garbled transcript	Low STT confidence score	Agent asks for a repeat rather than guessing entity values (especially `PTP_Amount`/`PTP_Date`)

8. Observability Metrics

Metric	Definition	Why It Matters
Containment Rate	% of calls resolved without	Measures automation

Metric	Definition	Why It Matters
	human escalation	efficiency; low rate signals prompt gaps or frequent hardship/dispute volume
PTP Rate	% of calls ending in a valid, logged Promise-to-Pay	Primary business outcome metric for collections effectiveness
First Call Resolution (FCR)	% of calls ending with a valid, non-ambiguous disposition logged	Tracks whether the agent reliably reaches a clean end-state rather than dropping/erroring out
Auth Success Rate	% of `verify_customer` calls returning `verified: true` on first attempt	Signals whether verification UX (prompt wording) is clear
Average Handle Time (AHT)	Mean call duration from `INIT` to `CALL_ENDED`	Efficiency and customer-experience proxy
Compliance Violation Count	Instances where debt terms were used before `AUTHENTICATED`	Should be zero; tracked via automated transcript scanning (see `tests/test_cases.json` TC-001)

Appendix: Sequence Diagram (Mermaid.js)

```
sequenceDiagram
    autonumber
    actor Customer
    participant Telephony as Telephony / SIP
    participant Vapi as Vapi Engine
    participant STT as Deepgram STT
    participant LLM as GPT-4o (Orchestrator)
    participant Server as Mock Webhook API
    participant TTS as ElevenLabs TTS

    Customer->>Telephony: Answers Call
    Telephony->>Vapi: Stream Audio
    Vapi->>STT: Real-time Audio Stream
    STT-->>Vapi: Transcribed Text Stream

    rect rgb(240, 240, 240)
        note over Vapi, LLM: Auth Phase (No Debt Disclosed)
        Vapi->>LLM: Send Conversation State + Transcript
        LLM-->>Vapi: Request Verification ("Provide last 4 digits of PAN")
        Vapi->>TTS: Synthesize Speech
        TTS-->>Customer: Play Audio
```

```
        Customer->>Vapi: Speaks ("1-2-3-4")
        Vapi->>LLM: Transcript ("1234")
        LLM->>Server: Tool Call: verify_customer(account_id, "1234")
        Server-->>LLM: Response: { verified: true, customer_name:
"Rahul Sharma" }
    end

    rect rgb(220, 245, 220)
        note over Vapi, LLM: Collections & Negotiation Phase
        LLM-->>Vapi: Disclose Debt & Ask PTP
        Vapi->>TTS: Audio Output ("₹8,499 overdue by 12 days...")
        TTS-->>Customer: Play Audio
        Customer->>Vapi: "I will pay this Friday."
        LLM->>Server: Tool Call: log_promise_to_pay(date: "2026-08-
14", amount: 8499)
        Server-->>LLM: Response: { status: "SUCCESS", ptp_id: "PTP-
9921" }
        LLM->>Server: Tool Call: send_payment_link(channel: "SMS")
        Server-->>LLM: Response: { link_sent: true }
    end

    LLM-->>Vapi: Final Polite Goodbye
    Vapi->>Customer: End Call
```

Render this diagram to `System_Architecture.png` using the Mermaid Live Editor (paste the code above, export as PNG) and drop the exported image into `docs/System_Architecture.png`.